Marwadi University	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP	**Aim:** Design Butterworth and Chebyshev filter using bilinear transformation method.	
Experiment-4	**Date: 24-08-26**	**Enrollment No:92400133128**

AIM: Design Butterworth and Chebyshev filter using bilinear transformation method.

Theory:

The bilinear transformation method is commonly used to design analog filters and then convert them into digital filters. This method maps the analog frequency response to the digital frequency response using a bilinear transformation.

The Butterworth and Chebyshev filters are two commonly used filter types. The Butterworth filter has a maximally flat frequency response in the passband, while the Chebyshev filter allows for a sharper transition between the passband and the stopband at the expense of ripples in either the passband or stopband.

1. Objectives

- To understand the concept of digital filters.
- To understand the working of Butterworth and Chebyshev filters.
- To design a low-pass Butterworth filter.
- To design a low-pass Chebyshev filter.

Chebyshev Filter

- Chebyshev filter provides a **sharper transition** between passband and stopband.
- It has ripple in the passband for the Type-I filter used in the program.
- The amount of ripple is controlled by the ripple value.
- In this experiment, the ripple value is **0.5 dB**.
- A **4th-order low-pass Chebyshev filter** is designed.

Bilinear Transformation

The **bilinear transformation method** is used to convert an analog filter into a digital filter.

The basic transformation is:

$$s = \frac{2}{T}\frac{1 - z^{-1}}{1 + z^{-1}}$$

where:

Marwadi University	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP	**Aim:** Design Butterworth and Chebyshev filter using bilinear transformation method.	
Experiment-4	**Date: 24-08-26**	**Enrollment No:92400133128**

- s= analog-domain variable
- z= digital-domain variable
- T= sampling period

In the given program, the bilinear() function performs this conversion.

Filter Specifications

Parameter	Value
Filter order	4
Cutoff frequency	1000 Hz
Sampling frequency	8000 Hz
Chebyshev ripple	0.5 dB

Program:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import butter, bilinear, freqz, cheby1

def design_butterworth_filter(filter_order, cutoff_frequency, sampling_frequency): #
  Design the analog Butterworth filter
  analog_b, analog_a = butter(filter_order, cutoff_frequency, analog=True, btype='low')

  # Perform the bilinear transformation
  digital_b, digital_a = bilinear(analog_b, analog_a,

  sampling_frequency) return digital_b, digital_a

def design_chebyshev_filter(filter_order, cutoff_frequency, sampling_frequency, ripple): #
  Design the analog Chebyshev filter
  analog_b, analog_a = cheby1(filter_order, ripple, cutoff_frequency, analog=True,
  btype='low')

  # Perform the bilinear transformation
  digital_b, digital_a = bilinear(analog_b, analog_a,
```

Marwadi University	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP	**Aim:** Design Butterworth and Chebyshev filter using bilinear transformation method.	
Experiment-4	**Date: 24-08-26**	**Enrollment No:92400133128**

```
    sampling_frequency) return digital_b, digital_a

def plot_filter_response(digital_b, digital_a, sampling_frequency): #
    Compute the frequency response of the filter
    frequency, magnitude_response = freqz(digital_b, digital_a, fs=sampling_frequency)

    # Plot the magnitude response
    plt.figure(figsize=(10, 6))
    plt.plot(frequency,
```

```
np.abs(magnitude_response))
plt.title('Filter Magnitude
Response') plt.xlabel('Frequency
(Hz)') plt.ylabel('Magnitude')
plt.grid(True) plt.show()

# Compute the impulse response of the filter
_, impulse_response = freqz(digital_b, digital_a, fs=sampling_frequency, worN=4096)

# Plot the impulse response
plt.figure(figsize=(10, 6))
plt.plot(impulse_response)
plt.title('Filter Impulse
Response') plt.xlabel('Samples')
plt.ylabel('Amplitude')
plt.grid(True) plt.show()

# Specify the desired filter specifications filter_order
= 4 # Filter order
cutoff_frequency = 1000 # Cutoff frequency in Hz
sampling_frequency = 8000 # Sampling frequency in
Hz ripple
= 0.5 # Ripple factor for Chebyshev filter

# Design the Butterworth filter
digital_b, digital_a = design_butterworth_filter(filter_order, cutoff_frequency,
sampling_frequency)

# Plot the Butterworth filter's magnitude response and
impulse response plot_filter_response(digital_b, digital_a,
sampling_frequency)
```

Marwadi University	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP	**Aim:** Design Butterworth and Chebyshev filter using bilinear transformation method.	
Experiment-4	**Date: 24-08-26**	**Enrollment No:92400133128**

```
# Design the Chebyshev filter
digital_b, digital_a = design_chebyshev_filter(filter_order, cutoff_frequency,
sampling_frequency, ripple)

# Plot the Chebyshev filter's magnitude response and impulse
response plot_filter_response(digital_b, digital_a,
sampling_frequency)

# Save the filter coefficients (optional) filter_path
= 'filter_coefficients.txt'
np.savetxt(filter_path, np.vstack((digital_b, digital_a)),
delimiter=',') print(f"Filter coefficients saved at: {filter_path}")
```

Marwadi University	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP	**Aim:** Design Butterworth and Chebyshev filter using bilinear transformation method.	
Experiment-4	**Date: 24-08-26**	**Enrollment No:92400133128**

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import butter, bilinear, freqz, cheby1


# Design Butterworth filter
def design_butterworth_filter(filter_order, cutoff_frequency, sampling_frequency):

    # Design the analog Butterworth filter
    analog_b, analog_a = butter(
        filter_order,
        cutoff_frequency,
        analog=True,
        btype='low'
    )

    # Perform the bilinear transformation
    digital_b, digital_a = bilinear(
        analog_b,
        analog_a,
        sampling_frequency
    )

    return digital_b, digital_a


# Design Chebyshev filter
def design_chebyshev_filter(
    filter_order,
    cutoff_frequency,
    sampling_frequency,
    ripple
):

    # Design the analog Chebyshev filter
    analog_b, analog_a = cheby1(
        filter_order,
        ripple,
        cutoff_frequency,
        analog=True,
        btype='low'
    )

    # Perform the bilinear transformation
    digital_b, digital_a = bilinear(
        analog_b,
        analog_a,
        sampling_frequency
    )

    return digital_b, digital_a

# Plot filter response
def plot_filter_response(
    digital_b,
    digital_a,
    sampling_frequency
):

    # Compute the frequency response
    frequency, magnitude_response = freqz(
        digital_b,
        digital_a,
        fs=sampling_frequency
    )

    # Plot magnitude response
    plt.figure(figsize=(10, 6))

    plt.plot(
        frequency,
        np.abs(magnitude_response)
    )

    plt.title('Filter Magnitude Response')
    plt.xlabel('Frequency (Hz)')
    plt.ylabel('Magnitude')
    plt.grid(True)
    plt.show()

    # Compute impulse response
    _, impulse_response = freqz(
        digital_b,
        digital_a,
        fs=sampling_frequency,
        worN=4096
    )

    # Plot impulse response
    plt.figure(figsize=(10, 6))

    plt.plot(impulse_response)

    plt.title('Filter Impulse Response')
    plt.xlabel('Samples')
    plt.ylabel('Amplitude')
    plt.grid(True)
    plt.show()
```

Marwadi University	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP	**Aim:** Design Butterworth and Chebyshev filter using bilinear transformation method.	
Experiment-4	**Date: 24-08-26**	**Enrollment No:92400133128**

```
# Specify filter specifications
filter_order = 4
cutoff_frequency = 1000
sampling_frequency = 8000
ripple = 0.5


# Design the Butterworth filter
digital_b, digital_a = design_butterworth_filter(
    filter_order,
    cutoff_frequency,
    sampling_frequency
)

# Plot Butterworth responses
plot_filter_response(
    digital_b,
    digital_a,
    sampling_frequency
)


# Design the Chebyshev filter
digital_b, digital_a = design_chebyshev_filter(
    filter_order,
    cutoff_frequency,
    sampling_frequency,
    ripple
)

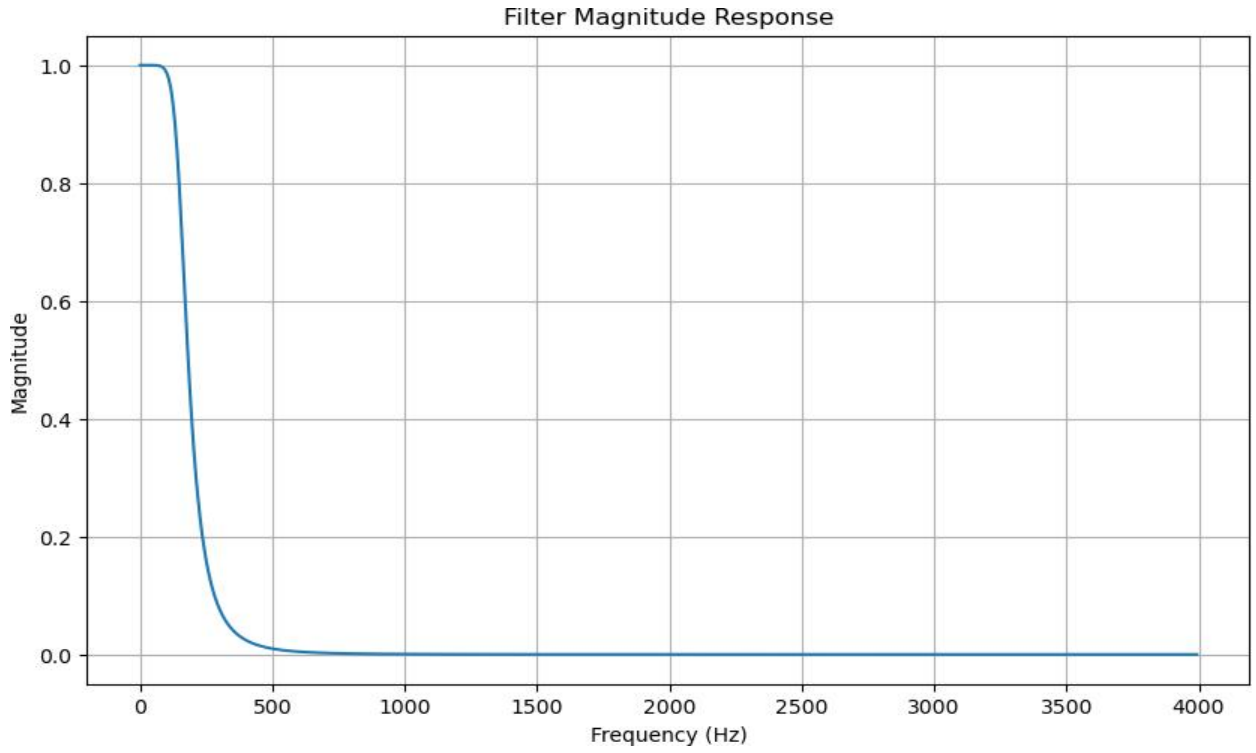
# Plot Chebyshev responses
plot_filter_response(
    digital_b,
    digital_a,
    sampling_frequency
)


# Save the filter coefficients
filter_path = 'filter_coefficients.txt'

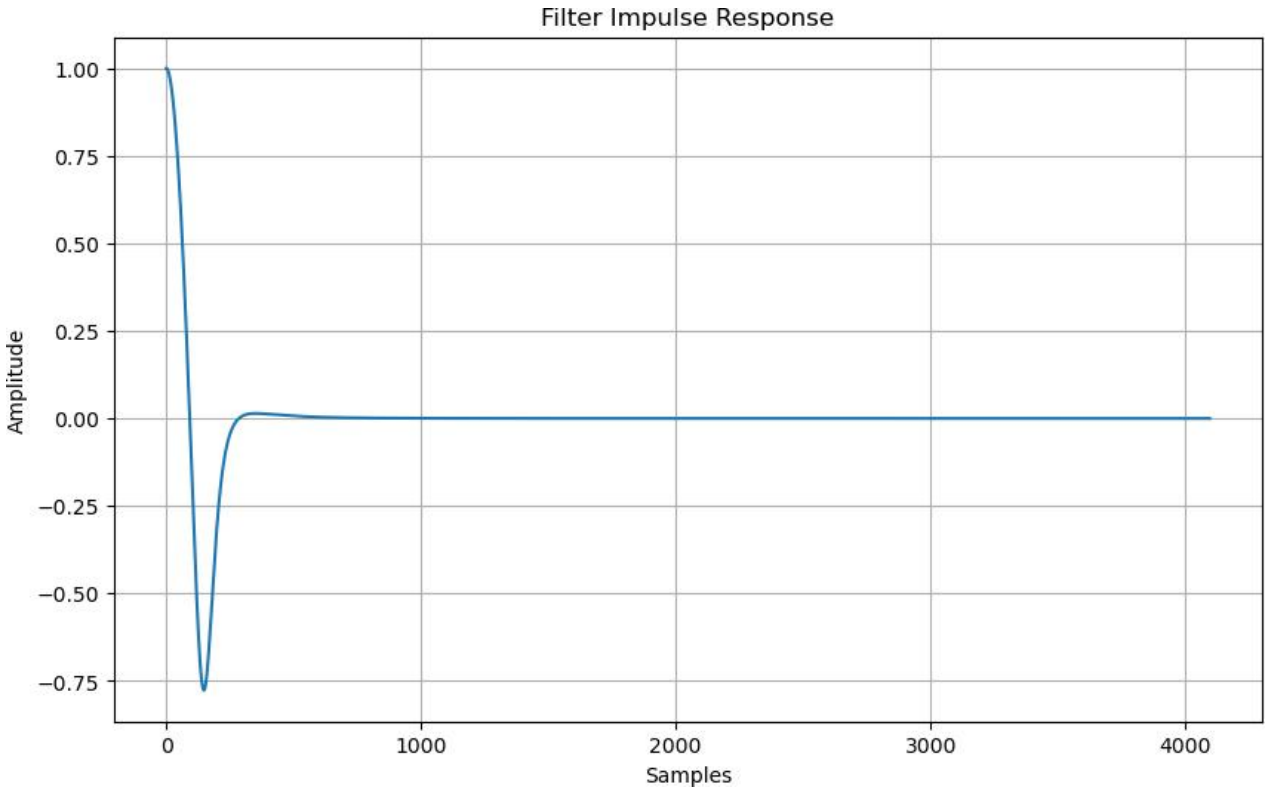
np.savetxt(
    filter_path,
    np.vstack((digital_b, digital_a)),
    delimiter=','
)

print(f"Filter coefficients saved at: {filter_path}")
```

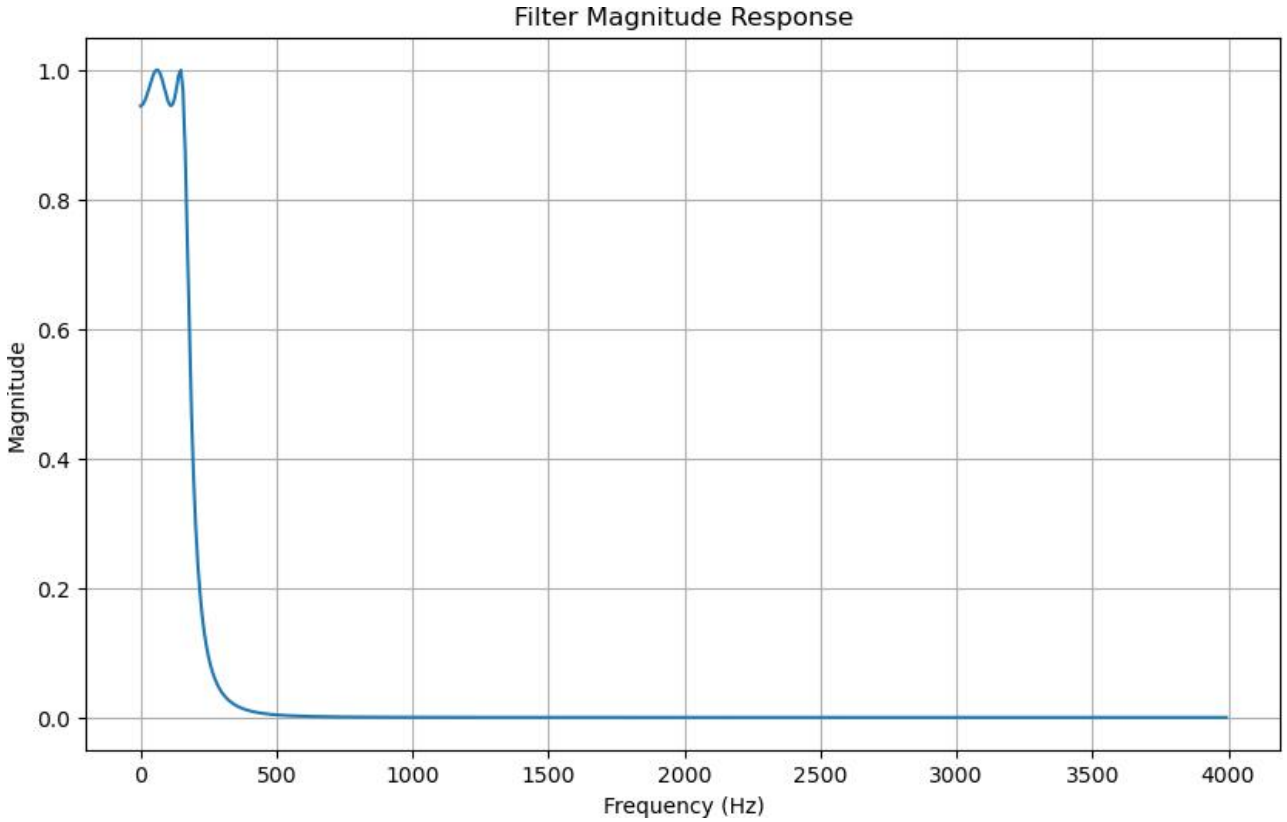
Marwadi University	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP	**Aim:** Design Butterworth and Chebyshev filter using bilinear transformation method.	
Experiment-4	**Date: 24-08-26**	**Enrollment No:92400133128**



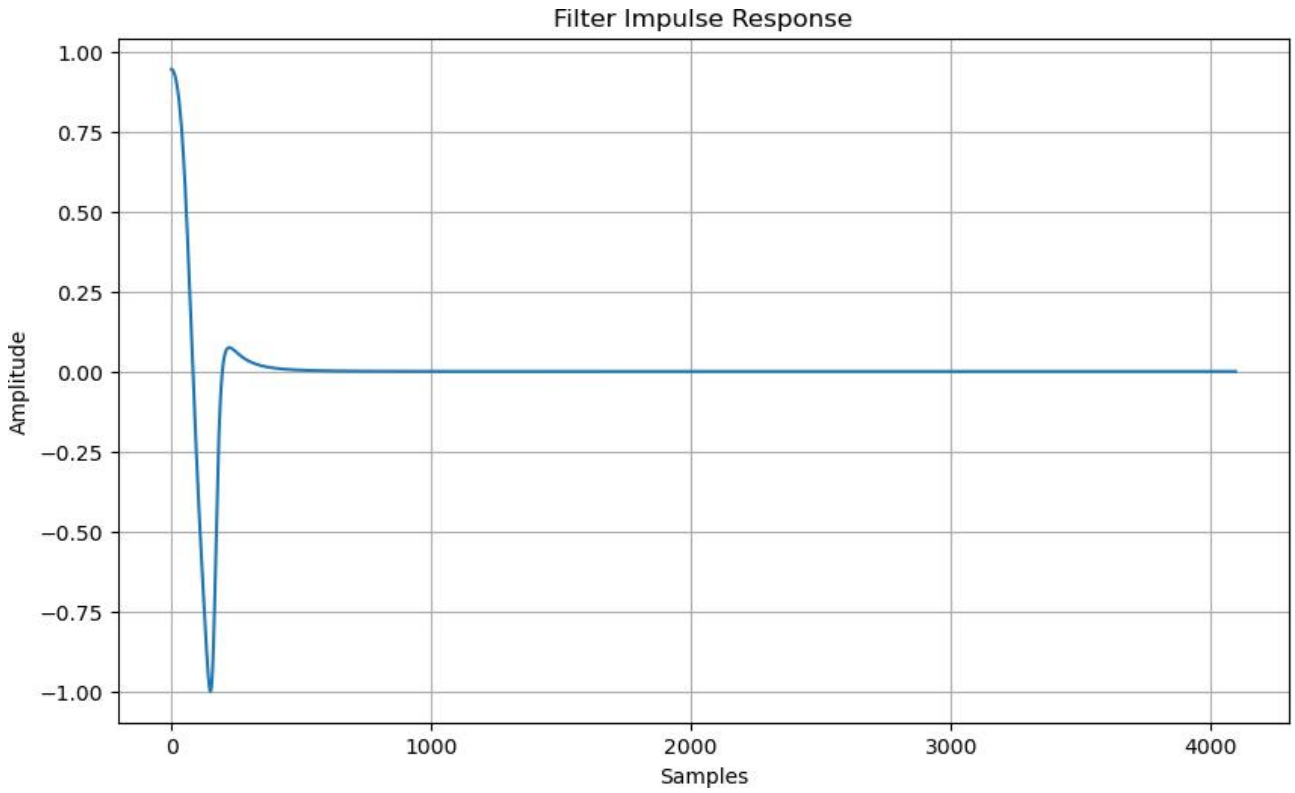
Marwadi University	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP	**Aim:** Design Butterworth and Chebyshev filter using bilinear transformation method.	
Experiment-4	**Date: 24-08-26**	**Enrollment No:92400133128**



Marwadi University	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP	**Aim:** Design Butterworth and Chebyshev filter using bilinear transformation method.	
Experiment-4	**Date: 24-08-26**	**Enrollment No:92400133128**



Marwadi University	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP	**Aim:** Design Butterworth and Chebyshev filter using bilinear transformation method.	
Experiment-4	**Date: 24-08-26**	**Enrollment No:92400133128**



Marwadi University	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP	**Aim:** Design Butterworth and Chebyshev filter using bilinear transformation method.	
Experiment-4	**Date: 24-08-26**	**Enrollment No:92400133128**

Observation:

Butterworth: The magnitude response is smooth and maximally flat in the passband.
Chebyshev: The magnitude response has ripple in the passband and provides a sharper transition between passband and stopband.

Result:

The Butterworth and Chebyshev filters were successfully designed using the bilinear transformation method. The magnitude response and impulse response of both digital filters were obtained using Python.

Conclusion:

The experiment demonstrated the design of Butterworth and Chebyshev filters using bilinear transformation. The Butterworth filter provides a maximally flat passband response, while the Chebyshev filter provides a sharper transition with passband ripple.

Github : https://github.com/abhinayreddy-7/DSIP.git